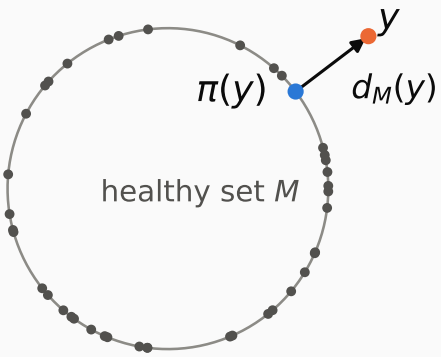
Gradient networks and the nearest point

Prop. 0.1, the max-of-planes picture, and the diffusion-geometry grad net

NP_AD

21 October 2026

The goal



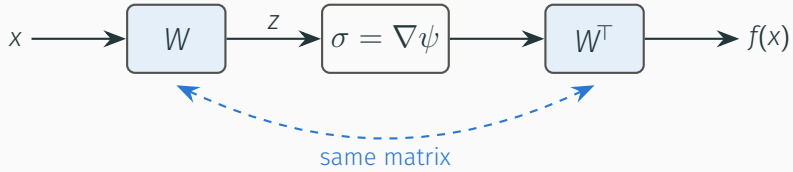
One scalar, four objects

$$F(y) = \tfrac{1}{2}\|y\|^2 - \tfrac{1}{2}d_M(y)^2$$

map	$\nabla F = \pi(y)$
distance potential	$\eta = \tfrac{1}{2}\|y\|^2 - F = \tfrac{1}{2}d_M^2$
residual	$\nabla\eta = y - \pi(y)$
score	$d_M = \sqrt{2\eta}$

Prop. 0.1: tied weights make a gradient

Prop. 0.1



$$W^\top \sigma(Wx + a) + b \;=\; \nabla_x \left[\psi(Wx + a) + b^\top x\right]$$

Read it as votes along the rows

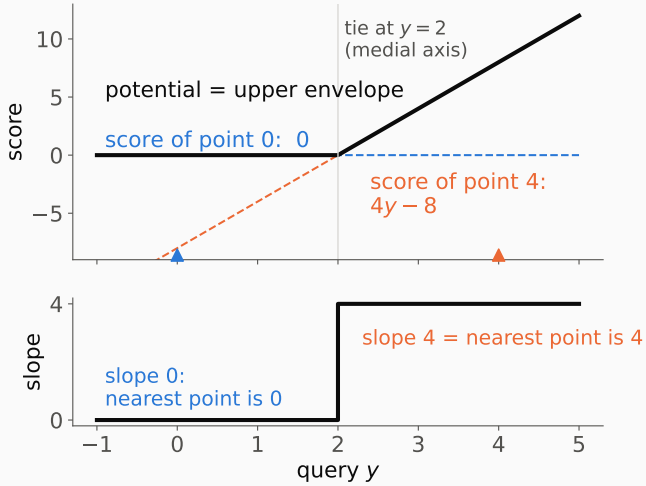
$$f(x) = \sum_i \sigma_i(z)\, w_i \;+\; b$$

Jacobian $= W^T J_\sigma W$ symmetric $\Rightarrow$ a gradient

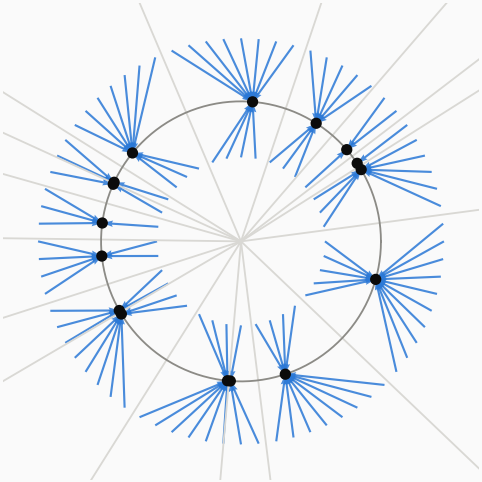
$J_\sigma \succeq 0 \;\Rightarrow$ convex potential, monotone map

Why the gradient is the nearest point

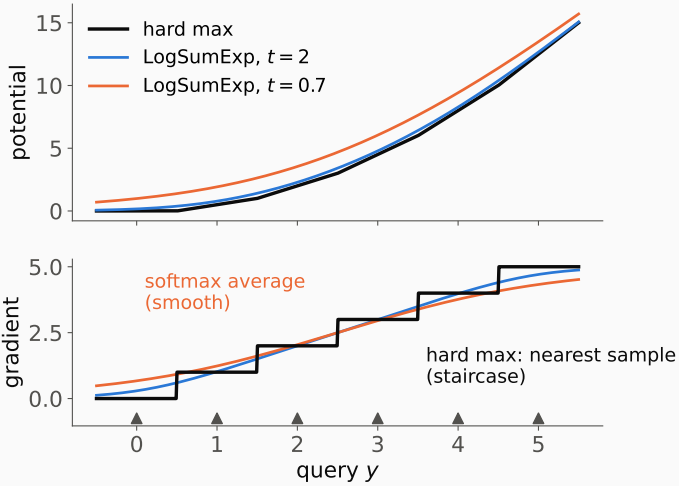
Nearest point = highest line



Many samples: one cell each



max → nearest sample; LogSumExp → smooth average



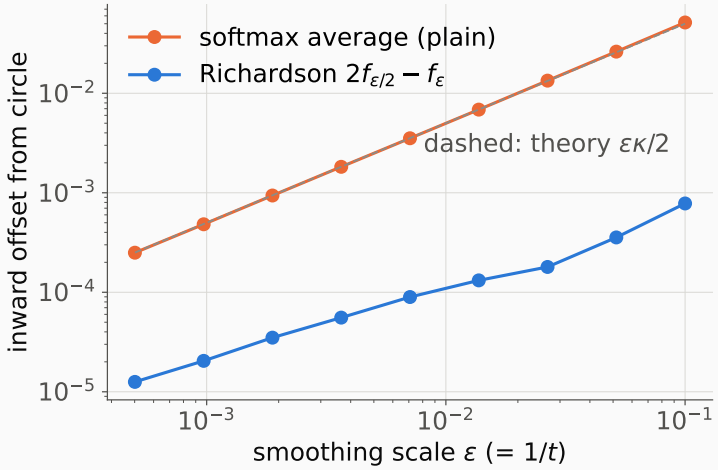
The kernel denoiser is this network

$$\text{rows} = \text{data} \qquad \text{offsets} = -\tfrac{1}{2}\|x_i\|^2 \qquad t = 1/\varepsilon$$

$$\Downarrow$$

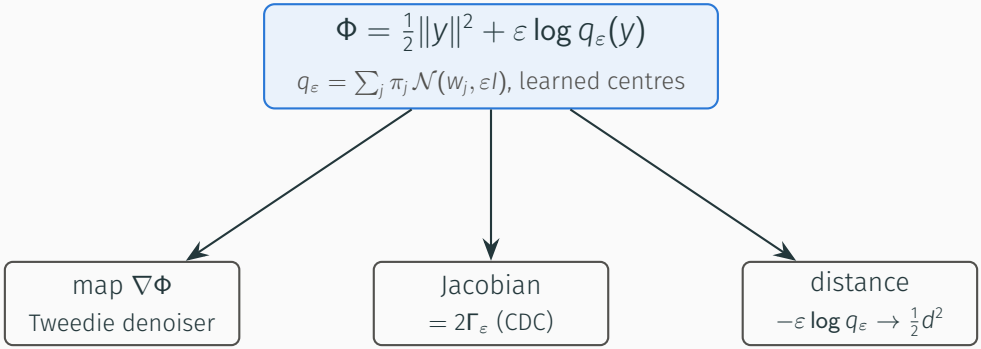
$$f(y) = \sum_i \mathrm{softmax}_i\left(-\tfrac{\|y-x_i\|^2}{2\varepsilon}\right) x_i$$

Smoothing costs a sag; two temperatures cancel it



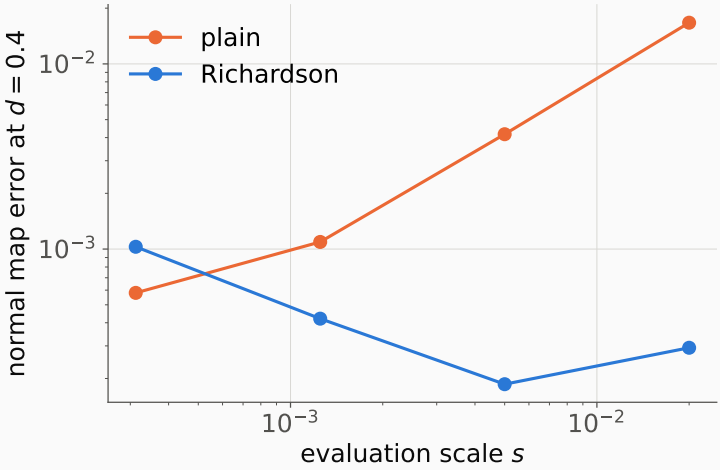
Our design

One network, three outputs, no eikonal

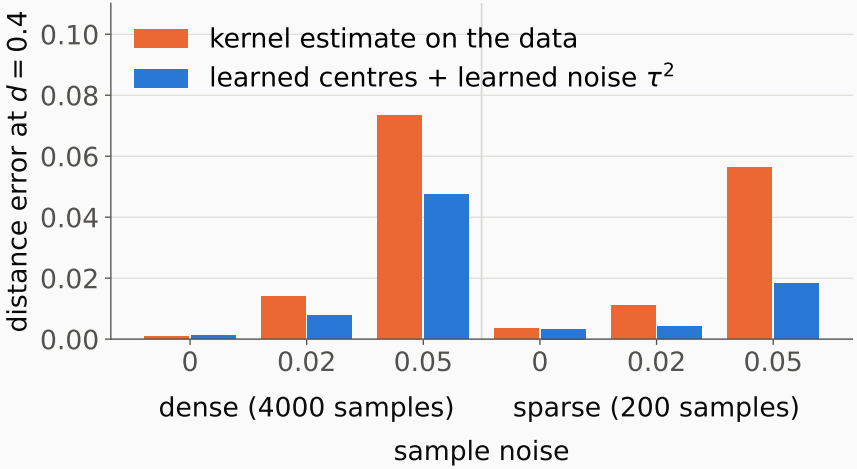


trained by denoising score matching only

Toy: Richardson removes the sag across scales

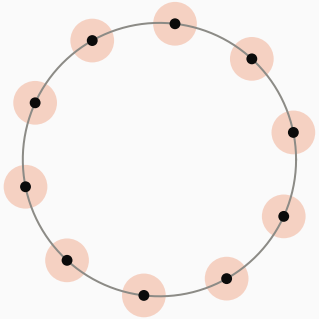


Toy: learning the noise helps when samples are noisy or sparse

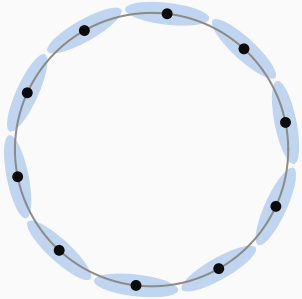


Next: stretch the bumps along the tangent

isotropic bumps (current)



stretched along the tangent



Where this stands

- toys only, one seed
- noise removal partial
- noise assumed isotropic
- evaluation scale: no rule yet
- real windows untested